

PrimeFocus

September 2025



WHEN

September 19, 2025

Doors open at 7:00pm

Meeting starts at 7:30pm

Lecture starts at 8:00pm

WHERE

Unitarian Church

1893 North Vasco Road

Livermore, CA 945571

and via Zoom

TVS QR CODE



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THIS MONTH'S SPEAKER

The Vera Rubin Observatory Don Dossa

After 20 years of proposals, multiple DOE and NSF reviews, the construction of the Vera Rubin Observatory on Cerro Pachon in Chile was finally approved. While a few minor construction details remain, the observatory has taken several weeks of test images.

This talk includes a review of the observatory, the technology, and its science goals. Early results from the Rubin will be presented. Differences with the James Webb Space Telescope (JWST) capabilities and operations are explained.



Rubin Observatory sits prominently atop Cerro Pachón against the backdrop of the full moon in this photo from March 24, 2024. The photo was taken around 7:55pm in the vicinity of Andacollo, Chile (about 25km from Rubin) after months of planning and patience. Credit, M.Hernández/H.Stockebrand

Don Dossa has a PhD in theoretical physics and most of his career has been in supercomputer design, computational astrophysics and astronomy. He has worked most recently at LLNL. Since retirement he has enjoyed amateur astronomy with his son Tom and wrote a novel.

NEWS AND NOTES

Meeting Dates

Club Meeting	Board Meeting	PrimeFocus Deadline
September 19	September 22	September 6
October 17	October 20	October 4
November 21	November 24	November 8

Money Matters

As of the last Treasurer’s Report on 8/18/25, our club’s account balance is \$45,998.15, this includes \$11,105.46 in the H2O Rebuild fund.

TVS Welcomes New Members

TVS welcomes new members Matthew Titus, Mazen Ataya, Daniel Dowling, Samuel Pollard, Justin Barry, Suhas Prakash, Ernest Epley, Jimin Kang, and Chandra Arumalla. Please say hello and chat with them during our meetings.

2025 TVS Club Star Party Schedule

Save the dates for the 2025 Club Star Parties. Del Valle star parties are also public outreach events. They are jointly hosted with the EBRPD and held at the Arroyo Staging Area (Coords: 37.6196638, -121.7528899). The public is invited for the first 1.5-2 hours, while club members can stay the remainder of the night.

No events currently scheduled for Del Valle.

Tesla Vintners star parties are open to only club members and their guests. These star parties end at midnight, but participants can leave earlier, should they wish.

No events currently schedule for Tesla Vintners

H2O Open House star parties are open to only club members and their guests. The open house ends at midnight, and all participants are encouraged to stay the duration. The drive to H2O takes about 1 hour, and the caravan leaves promptly from the corner of Mines and Tesla Rds. No gas stations are available on the route, so be prepared. Admission is \$3/car-bring exact change. H2O is a primitive site with two porta-potties. Bring water, food, and warm clothing, as needed. Red flashlights are to be used so observers can preserve their night vision.

No events currently scheduled for H2O
April 30: School star party at John Green Elementary, 3300 Antone Way, Dublin. Set-up 6:30pm, Observing from 7:30-9:30pm.

September 30: Public Star Party at Sunol Regional Wilderness, 1895 Geary Road, Sunol ([Google Map](#)). Set-up 6:45pm, Observing from 7:30-9:30pm.

October 1: Stargazing at the Livermore library, 1188 S Livermore

Ave, Livermore, CA. Set-up 6:30pm, Observing from 7:00-9:30pm

October 6: School star party at John Green Elementary, 3300 Antone Way, Dublin, CA. Set-up 6:30pm, Observing from 7:30-9:00pm

October 10: School star party at Murray Elementary, 8435 Davona Dr, Dublin, CA 94568. Set-up 6:30pm, Observing from 7:15-9:30pm



Images from Del Valle Star Party 2025, Images provided by Johnathan Bailey and Craig Siders

CALENDAR OF EVENTS



September 19, 20, 26, 27, October 3, 4, 10, 11, 17, 18, 7:30-10:30pm

What	Free Telescope Viewing
Who	Chabot Staff
Where	Chabot Space and Science Center, 10000 Skyline Blvd. Oakland, CA 94619
Cost	Free

Join Chabot astronomers on the Observatory Deck for a free telescope viewing! Weather permitting, this is a chance to explore stars, planets and more through Chabot's historic telescopes. Chabot's three large historic telescopes offer a unique way to experience the awe and wonder of the Universe. Three observatory domes house the Center's 8-inch (Leah, 1883) and 20-inch (Rachel, 1916) refracting telescopes, along with a 36-inch reflecting telescope (Nellie, 2003).

Are the skies clear for viewing tonight? Viewing can be impacted by rain, clouds, humidity and other weather conditions. Conditions can be unique to Chabot because of its unique location in Joaquin Miller Park. Before your visit, check out the Weather Station to see the current conditions at Chabot.

For more information, see
<https://chabotspace.org/events/events-listing/>
<https://eastbayastro.org/chabot-telescope-status/>

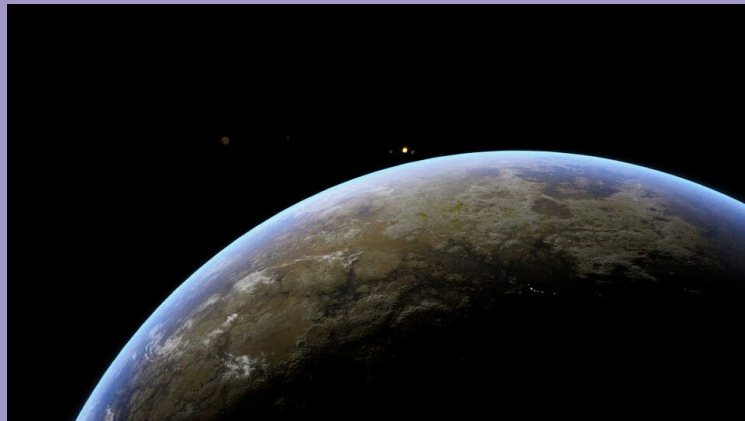
October 6, 7:30 PM

What	The Search for Life in the Universe
Who	California Academy of Sciences
Where	Morrison Planetarium; 55 Music Concourse Drive, San Francisco, CA 94118
Cost	Public: \$15; Members and seniors: \$12

Featuring Giada Arney, NASA's Goddard Space Flight Center

Humans have wondered whether we are alone in the universe for millennia. Finally, a future NASA observatory may bring us the answers. NASA's upcoming Habitable Worlds Observatory will follow in the tradition of other great observatories like Hubble and the James Webb Space Telescope. It will be the most powerful telescope NASA has ever launched. Excitingly, it

will be the first facility designed specifically to search for signs of habitability and life on planets outside of our solar system. But correctly decoding the signatures of life on alien worlds will take more than just a powerful tool. Join us as we discuss strategies for life detection on worlds beyond our solar system for the next era of cosmic observations.



Are there other Earth-like worlds beyond our solar system? Are we alone in the universe? A future NASA mission will seek the answers.

Giada Arney is an astrobiologist at NASA's Goddard Space Flight Center. She serves as interim Project Scientist for NASA's future Habitable Worlds Observatory and as Deputy Principal Investigator for NASA's DAVINCI mission to Venus. She received a B.A. in astronomy from the University of Colorado at Boulder and a PhD in astronomy and astrobiology from the University of Washington. After joining NASA in 2016, she received the NASA Early Career Achievement Medal in 2018 and was a recipient of the Presidential Early Career Award for Scientists and Engineers in 2019. She loves planetariums and has enjoyed the privilege of working at Fiske Planetarium at the University of Colorado as an undergraduate student and was planetarium coordinator for the University of Washington for two years during graduate school.



October 8, 7:00 PM

What	The New Vera C. Rubin Observatory: Surveying the Universe
Who	Silicon Valley Astronomy Lecture Series
Where	Smithwick Theater at Foothill College, in Los Altos, https://foothill.edu/map/
Cost	Free

The Vera C. Rubin Observatory is a unique ground-based astronomy facility, with the largest digital camera in the world, designed to provide a time-lapse “movie” of the entire sky from the Earth’s southern hemisphere. Over its planned ten years of operation, the Rubin Observatory will obtain nearly 1,000 images of every part of that sky. By comparing the various images, we will be able to detect everything that varies in brightness and everything that moves across the sky. By adding together all of the images, we will be able to catalog nearly 20 billion galaxies and a comparable number of stars. After 20 years of development, this facility has just come on-line and will soon begin its nightly operations. Prof. Kahn will review the design, development, and construction of Rubin, and describe the exciting science that lies ahead.



Rubin Observatory stands on Cerro Pachón in Chile against a sky full of star trails in this long exposure night sky image. Credit: Hernan Stockebrand

Steven M. Kahn is the Dean of Mathematical and Physical Sciences and a Professor of Physics and Astronomy at the University of California, Berkeley. Prior to returning to Berkeley in 2022, Kahn was the Cassius Lamb Kirk Professor in the Natural Sciences at Stanford, and the I.I. Rabi Professor of Physics and Astronomy at Columbia. Kahn received his A.B. degree from Columbia and Ph.D. in physics from Berkeley before beginning his academic career. Kahn is an experimental astrophysicist and cosmologist who has made major contributions to multiple fields. He was the U.S. Principal Investigator on the Reflection Grating Spectrometer experiment launched on the European XMM- Newton Observatory in 1999. More recently, he served as the Director of the Rubin Observatory construction project from its inception in 2013 until his move to Berkeley in 2022.

Kahn is an elected Fellow of the American Academy of Arts and Sciences, and an elected Member of the National Academy of Sciences.

Foothill College is just off the El Monte Road exit from Freeway 280 in Los Altos.

For directions and parking information, see: <https://foothill.edu/parking/>

For a campus map, to find the Smithwick Theater (Bldg. 1000), see:

<https://foothill.edu/map/>

Note: Parking lot 1 is closest, with access to the theater by stairs. Parking lot 5 provides access from the same elevation as the theater.

The lecture is co-sponsored by:

*The Foothill College Science, Tech, Engineering & Math Division

*The SETI Institute and

*The Astronomical Society of the Pacific

Past lectures in the series can also be found on YouTube at:

<http://youtube.com/svastronomylectures>

and as audio podcasts at: <https://www.buzzsprout.com/1805595>



SH2-155 The Cave Nebula, By Mark Hai Du.
For full size image, see <https://app.astrobin.com/i/mntrg0>



NGC 7331, By Swaroop Shere.

For full size image, see <https://app.astrobin.com/i/16a78x?r=C>



SN 2025RBS (NGC 7331), By Swaroop Shere.

For full size image, see <https://app.astrobin.com/i/16a78x?r=0>

WHATS UP

Adapted from Sky and Telescope

All times are Pacific Standard Time

September 2025

20-21	Sat-Sun	All night see Saturn at opposition
21	Sun	New Moon
22	Mon	Autumn begins at 11:19AM
27	Sat	At dusk moon is just 3° just left of Antares
29	Mon	Moon is at first quarter

October 2025

5	Sun	At dusk facing East Southeast, see the Moon about 2° upper left of Saturn
7	Tue	New Moon
9	Thu	Evening look for the Moon rising with the Pleiades above the Eastern-Northeastern horizon
12	Sun	Algol shines a minimum brightness between 8:42pm and 10:42pm
13	Mon	Moon is at last quarter
14	Tue	See the Moon, Jupiter, & Pollux form an equilateral triangle with the objects just 6½° apart
16	Thu	In the morning the moon leads Regulus by 4° while it rises in the East
19	Sun	At Dawn the Moon rises just 4° upper right of Venus

OFFICERS AND VOLUNTEER POSITIONS

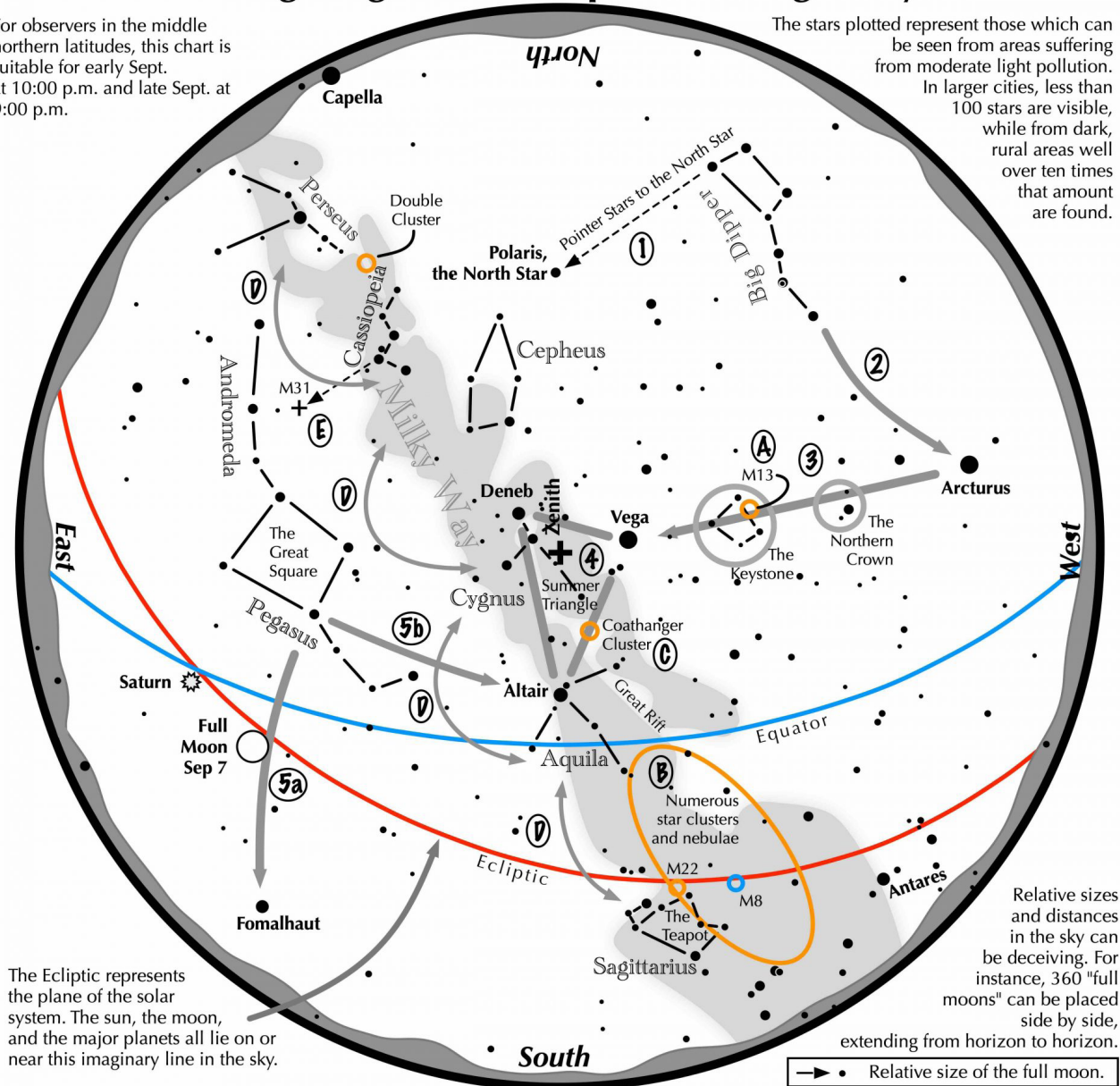
Officers	Volunteer Positions	Newsletter	Potluck Coordinator
President Eric Dueltgen president@trivalleystargazers.org	Astronomical League Rep. Don Dossa alrep@trivalleystargazers.org	Scott Schneider (Editor) newsletter@trivalleystargazers.org	OPEN potluck@trivalleystargazers.org
Vice-President Aris Pope vice_president@trivalleystargazers.org	Club Star Party Coordinator Johnathan Bailey coordinator@trivalleystargazers.org	Webmaster Swaroop Shere webmaster@trivalleystargazers.org	Program Coordinator Ron Kane programs@trivalleystargazers.org
Treasurer John Forrest treasurer@trivalleystargazers.org	Del Valle Coordinator David Wright delvalle@trivalleystargazers.org	Night Sky Network Rep. Ross Gaunt nnsn@trivalleystargazers.org	Publicity and Fundraising OPEN publicity@trivalleystargazers.org
Secretary Dave Lackey secretary@trivalleystargazers.org	Historian OPEN historian@trivalleystargazers.org	H2O Observatory Director / Rebuild Chairman Chuck Grant H2O@trivalleystargazers.org	Refreshment Coordinator OPEN
Past President Ron Kane past_president@trivalleystargazers.org	Librarian Ron Kane librarian@trivalleystargazers.org	Observing Program Coordinator Ron Kane awards@trivalleystargazers.org	Web and Email www.trivalleystargazers.org info@trivalleystargazers.org
	Loaner Scope Manager Ron Kane telescopes@trivalleystargazers.org	Outreach Coordinator Johnathan Bailey outreach@trivalleystargazers.org	TVS E-Group To Join the TVS E-Group just send an email to TVS at info@trivalleystargazers.org asking to join the group. Make sure you specify the email address you want to use to read and post to the group.

NAVIGATING THE NIGHT SKY FOR SEPTEMBER

Navigating the mid September Night Sky

For observers in the middle northern latitudes, this chart is suitable for early Sept. at 10:00 p.m. and late Sept. at 9:00 p.m.

The stars plotted represent those which can be seen from areas suffering from moderate light pollution. In larger cities, less than 100 stars are visible, while from dark, rural areas well over ten times that amount are found.



The Ecliptic represents the plane of the solar system. The sun, the moon, and the major planets all lie on or near this imaginary line in the sky.

Relative sizes and distances in the sky can be deceiving. For instance, 360 "full moons" can be placed side by side, extending from horizon to horizon.

→ • Relative size of the full moon.

Navigating the mid September night sky: Simply start with what you know or with what you can easily find.

- 1 Extend a line north from the two stars at the tip of the Big Dipper's bowl. It passes by Polaris, the North Star.
- 2 Follow the arc of the Dipper's handle. It intersects Arcturus, the brightest star in the September evening sky.
- 3 Nearly overhead shines a star of similar brightness as Arcturus, Vega. Draw a line from Arcturus to Vega. It first meets "The Northern Crown," then the "Keystone of Hercules." A dark sky is needed to see these two dim stellar configurations.
- 4 The stars of the summer triangle, Vega, Altair, and Deneb, shine overhead.
- 5 The westernmost two stars of the Great Square, which lies high in the east, point south to Fomalhaut. The southernmost two stars point west to Altair.

Binocular Highlights

- A: On the western side of the Keystone glows the Great Hercules Cluster.
- B: Between the bright stars Antares and Altair, hides an area containing many star clusters and nebulae.
- C: 40% of the way between Altair and Vega, twinkles the "Coathanger," a group of stars outlining a coathanger.
- D: Sweep along the Milky Way for an astounding number of faint glows and dark bays, including the Great Rift.
- E: The three westernmost stars of Cassiopeia's "W" point south to M31, the Andromeda Galaxy, a "fuzzy" oval.

Astronomical League www.astroleague.org/outreach; duplication is allowed and encouraged for all free distribution.



NASA NIGHT SKY NOTES

The NASA Night Sky Network (NSN) has been a leader in supporting astronomy clubs with club coordination and education materials for more than 21 years. We are proud to be one of the communities supported into the future of NASA, as cuts and restructuring are part of the Fiscal Year 2026 budget. Through your efforts, more than 8.4 million people have been introduced to the cosmos right in their own communities. We will continue to support clubs with much of what you have grown to know and love about NSN; however, as of October 1, 2025, Night Sky Network will be operating with the following updates:

Night Sky Notes will be suspended until further notice.



This article is distributed by NASA's Night Sky Network (NSN).

The NSN program supports astronomy clubs across the USA dedicated to astronomy outreach. Visit nightsky.jpl.nasa.gov to find local clubs, events, and more!



Tri-Valley Stargazers
P.O. Box 2476
Livermore, CA 94551
www.trivalleystargazers.org

Tri-Valley Stargazers Membership Application

Contact information:

Name: _____ Phone: _____

Street Address: _____

City, State, Zip: _____

Email Address: _____

Status (select one): _____ New member _____ Renewing or returning member

Membership category (select one): Membership term is for one calendar year, January through December.

_____ Student member (\$10). Must be a full-time high-school or college student.

_____ Regular member (\$30).

Hidden Hill Observatory Access (optional): Must be 18 or older.

_____ One-time key deposit (\$20). This is a refundable deposit for a key to H2O. New key holders must first hear an orientation lecture and sign a usage agreement form before using the observing site.

_____ Annual access fee (\$10). You must also be a key holder to access the site.

Donation (optional):

_____ Tax-deductible contribution to Tri-Valley Stargazers

Total enclosed: \$ _____

Member agrees to hold Tri-Valley Stargazers, and any cooperating organizations or landowners, harmless from all claims of liability for any injury or loss sustained at a TVS function. TVS will not share information with anyone except as detailed in our Privacy Policy (<http://www.trivalleystargazers.org/privacy.shtml>).

Mail this completed form along with a check to: Tri-Valley Stargazers, P.O. Box 2476, Livermore, CA 94551.